

Exp No. 12: NB-IoT vs LTE-M Throughput Simulation Date

AIM: To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

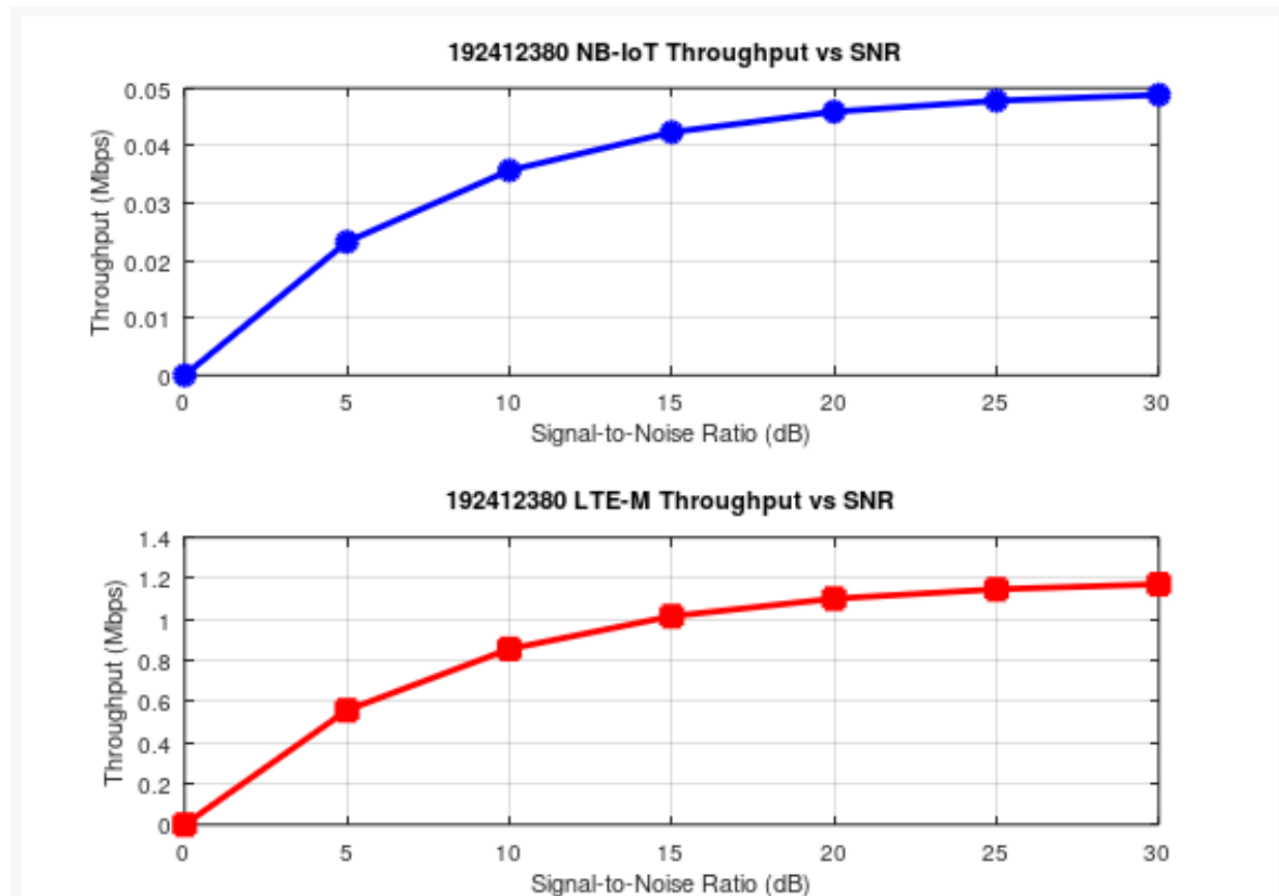
Objective 1:

To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % SNR values (dB)
6  SNR = 0:5:30;
7
8  % Simulated Throughput (Mbps)
9  NB_IoT = 0.05 * (1 - exp(-SNR/8));
10 LTE_M = 1.2 * (1 - exp(-SNR/8));
11
12 % Display results
13 disp('SNR(dB)      NB-IoT(Mbps)      LTE-M(Mbps)');
14 disp([SNR' NB_IoT' LTE_M']);
15
16 % Create figure
17 figure;
18
19 % ----- Graph 1: NB-IoT -----
20 subplot(2,1,1);
21 plot(SNR, NB_IoT, '-o', ...
22      'LineWidth', 2, ...
23      'Color', 'b', ...
24      'MarkerFaceColor', 'b');
25
26 title(' 192412380 NB-IoT Throughput vs SNR');
27 xlabel('Signal-to-Noise Ratio (dB)');
28 ylabel('Throughput (Mbps)');
29 grid on;
30
31 % ----- Graph 2: LTE-M -----
32 subplot(2,1,2);
33 plot(SNR, LTE_M, '-s', ...
34      'LineWidth', 2, ...
```

OUTPUT :



Objective 2:

To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

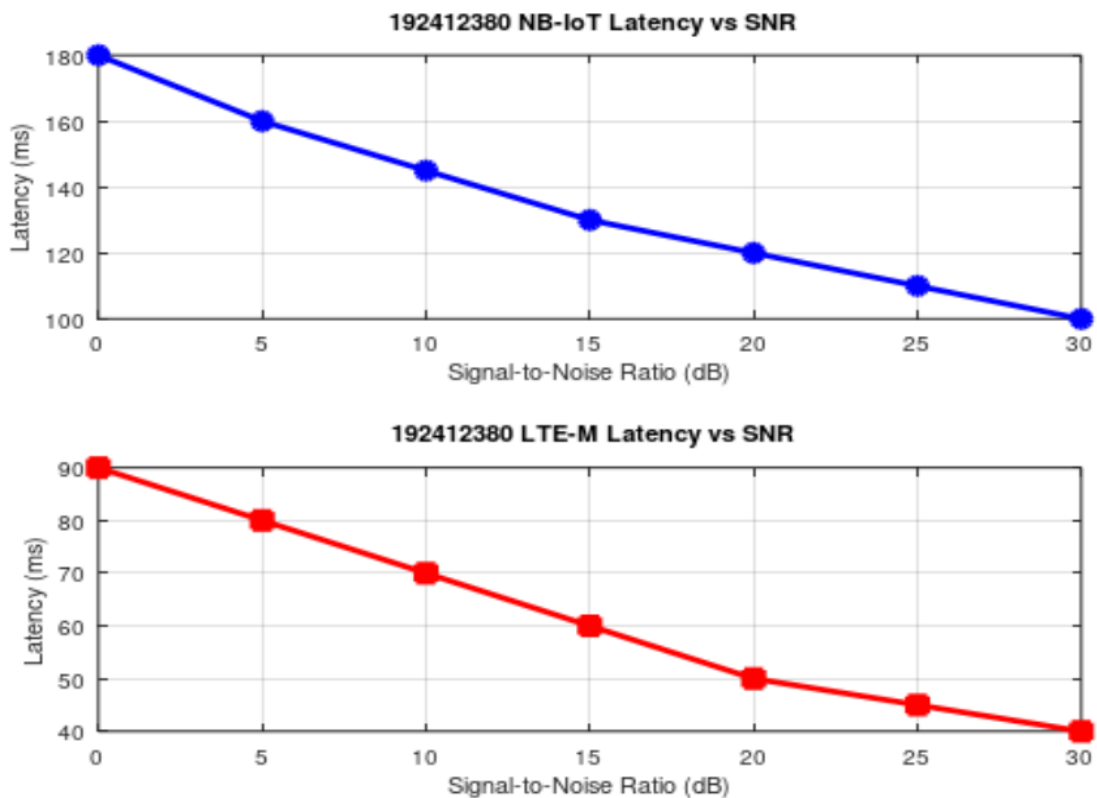
Program

```

1  clc;
2  clear;
3  close all;
4
5  % SNR values (dB)
6  SNR = 0:5:30;
7
8  % Simulated Latency (ms)
9  NB_IoT = [180 160 145 130 120 110 100];
10 LTE_M = [90 80 70 60 50 45 40];
11
12 % Display results
13 disp('SNR(dB)      NB-IoT Latency(ms)      LTE-M Latency(ms)');
14 disp([SNR' NB_IoT' LTE_M]);
15
16 % Create figure
17 figure;
18
19 % ----- Graph 1: NB-IoT -----
20 subplot(2,1,1);
21 plot(SNR, NB_IoT, '-o', ...
22      'LineWidth', 2, ...
23      'Color', 'b', ...
24      'MarkerFaceColor', 'b');
25
26 title('192412380 NB-IoT Latency vs SNR');
27 xlabel('Signal-to-Noise Ratio (dB)');
28 ylabel('Latency (ms)');
29 grid on;
30
31 % ----- Graph 2: LTE-M -----
32 subplot(2,1,2);
33 plot(SNR, LTE_M, '-s', ...
34      'LineWidth', 2, ...

```

OUTPUT :



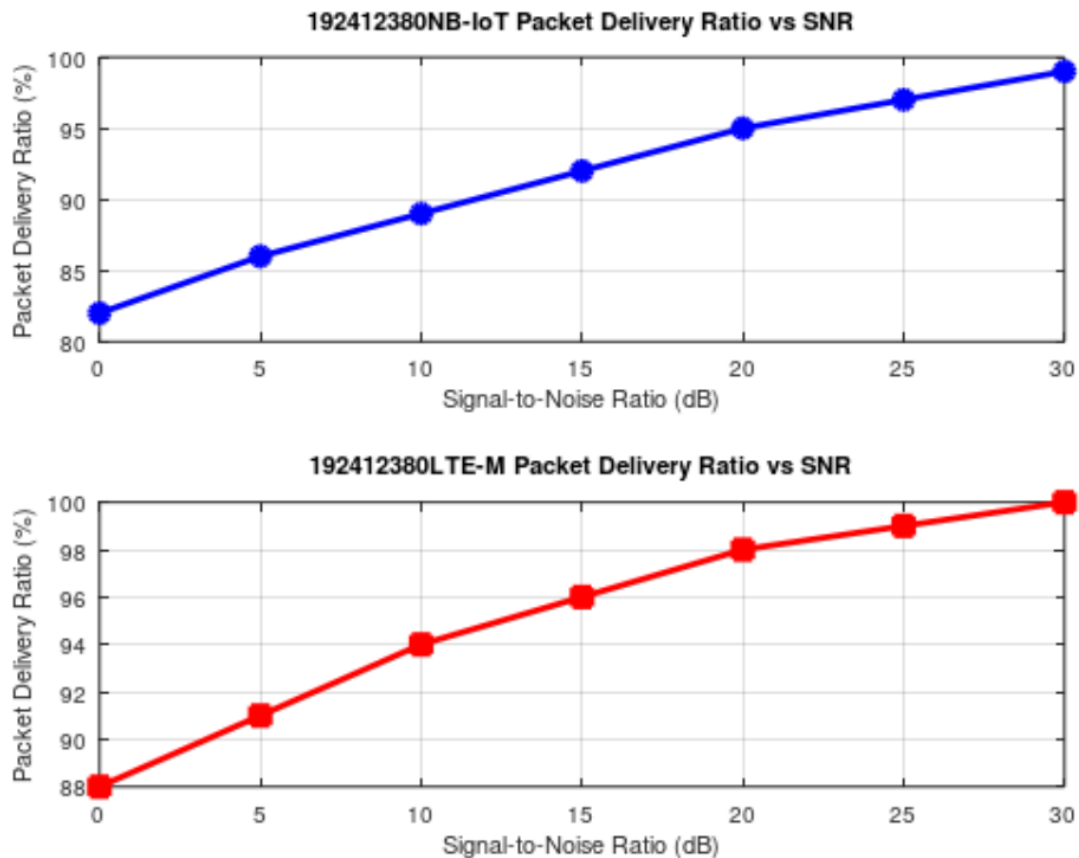
Objective 3:

To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

program

```
1  clc;
2  clear;
3  close all;
4
5  % SNR values (dB)
6  SNR = 0:5:30;
7
8  % Simulated Packet Delivery Ratio (%)
9  NB_IoT = [82 86 89 92 95 97 99];
10 LTE_M = [88 91 94 96 98 99 100];
11
12 % Display results
13 disp('SNR(dB)      NB-IoT PDR(%)      LTE-M PDR(%)');
14 disp([SNR' NB_IoT' LTE_M']);
15
16 % Create figure
17 figure(1);
18
19 % ----- Graph 1: NB-IoT -----
20 subplot(2,1,1);
21 plot(SNR, NB_IoT, '-ob', 'LineWidth', 2, 'MarkerFaceColor', 'b');
22 title('192412380NB-IoT Packet Delivery Ratio vs SNR');
23 xlabel('Signal-to-Noise Ratio (dB)');
24 ylabel('Packet Delivery Ratio (%)');
25 grid on;
26
27 % ----- Graph 2: LTE-M -----
28 subplot(2,1,2);
29 plot(SNR, LTE_M, '-sr', 'LineWidth', 2, 'MarkerFaceColor', 'r');
30 title('192412380LTE-M Packet Delivery Ratio vs SNR');
31 xlabel('Signal-to-Noise Ratio (dB)');
32 ylabel('Packet Delivery Ratio (%)');
33 grid on;
34
```

OUTPUT :



Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.
2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.
3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission.

Practical Questions

An IoT communication network requires reliable and efficient wireless technologies for connecting smart devices. Using MATLAB, simulate and compare the performance of **NB-IoT** and **LTE-M** in terms of throughput, latency, and packet delivery ratio.

1. To evaluate and compare the **Throughput (Mbps)** performance of NB-IoT and LTE-M under different network conditions.
2. To analyze the **Latency (ms)** characteristics of NB-IoT and LTE-M and study their impact on communication performance.
3. To measure and compare the **Packet Delivery Ratio (PDR %)** of NB-IoT and LTE-M systems for reliable data transmission.